HackAI 2026

Early Fault Detection *for use in Jet Engines*

Team: NaN

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Motivation

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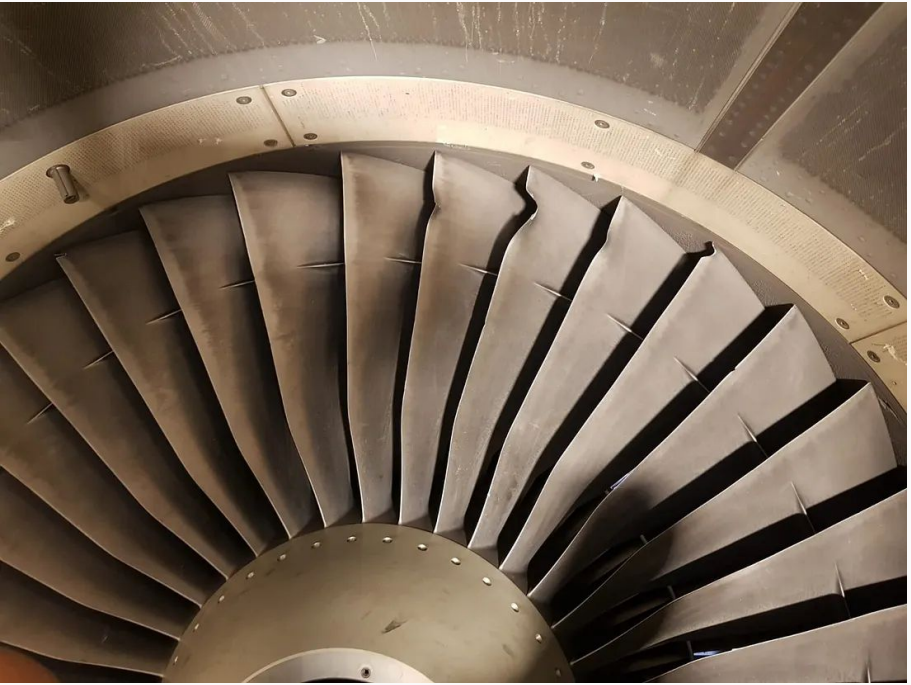

Oftentimes commercial aircraft engines face damages during flight, commonly known as "faults".

Faults can range from things like microscopic cracks in fan blades, to bigger dents, to volcanic ash coating the compressor, and can often be silent.

Airplane engines can operate for many cycles after the initial fault, and it is common for this initial fault to go unnoticed for many flights after it's initial occurrence.

According to industry estimates, unplanned downtime costs the global aviation sector more than $33 billion a year. Early fault detection can help cut these costs, and help reduce unplanned downtime!

https://veryon.com/blog/how-predictive-maintenance-is-revolutionizing-aircraft-reliability#:~:text=In%20aviation%2C%20a%20grounded%20aircraft,can%20disrupt%20the%20entire%20day.

The Dataset

CMAPSS Jet Engine Simulated Data from NASA:

- A collection of simulated data of jet engines with sensors.
  - The training data goes until the engine stops working entirely.
  - The testing data may or may not go until the engine stops working entirely. The engine may or may not even develop a fault in the provided time span.
  - The data was generated under hyper-realistic conditions utilizing NASA's official testing software, which accurately represent jet engine sensors, degradation, and real-world physics..
- We worked with two specific subsets: FD001 (idealistic) and FD004 (most realistic).

https://data.nasa.gov/dataset/cmapss-jet-engine-simulated-data

Handling the Data

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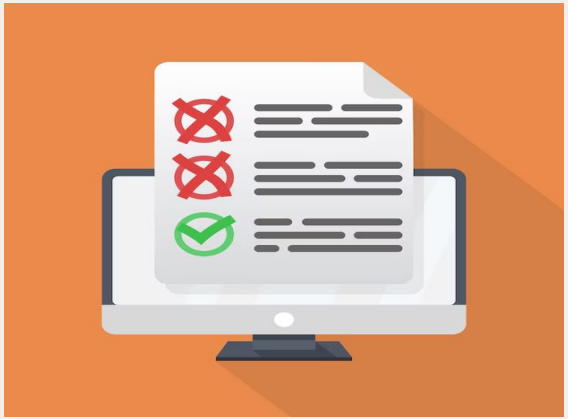
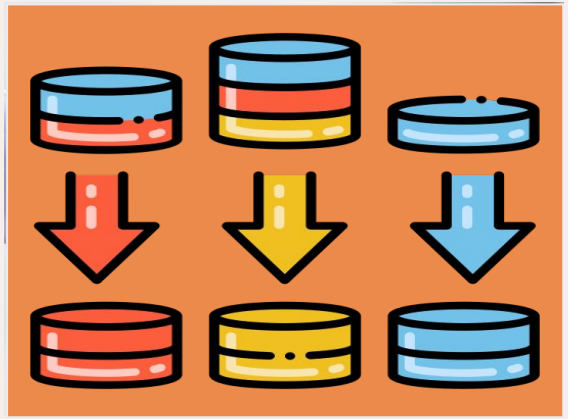


REMOVING SENSORS

- Removing sensors with constant data.
- Removing sensors that only have binary data (like an on/off switch), with no fluctuation over cycle time.

TRIMMING DEGRADATION DATA

- The autoencoder works by reconstructing pre-fault behavior
- Trimming data that occurs post-fault allows the autoencoder to “learn” only on pre-fault data



NORMALIZATION

- Normalizing the sensor data for data stability.
- Normalizing per flight condition regime, due to differences in relative scaling per regime.

The Models - LSTM

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- All models were unsupervised LSTM autoencoders, chosen to leverage the temporal nature of the CMAPSS dataset.
- We compared three models:
  - A baseline LSTM.
  - A K-clustered LSTM.
  - An attention augmented LSTM.

Baseline

For FD001

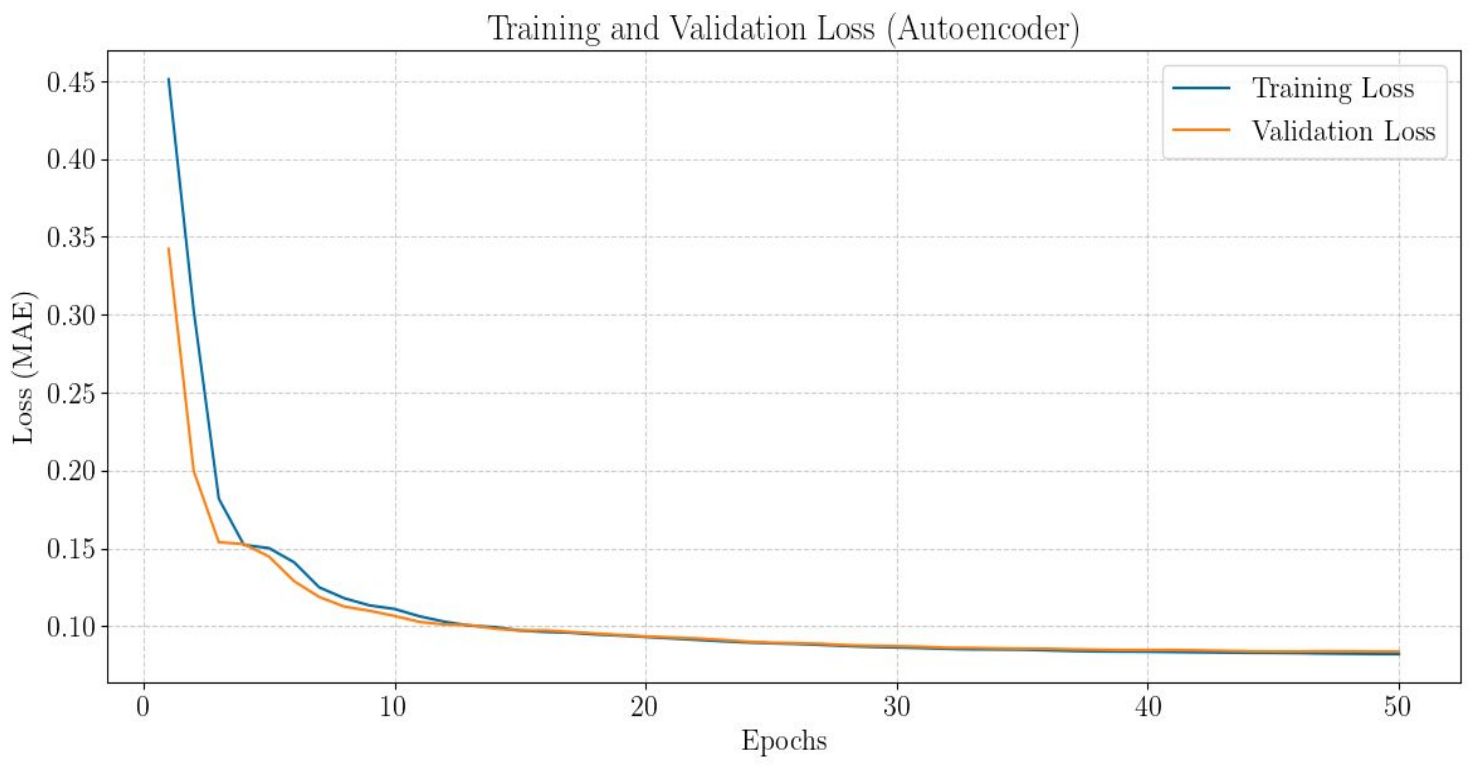
K-Cluster

For FD004

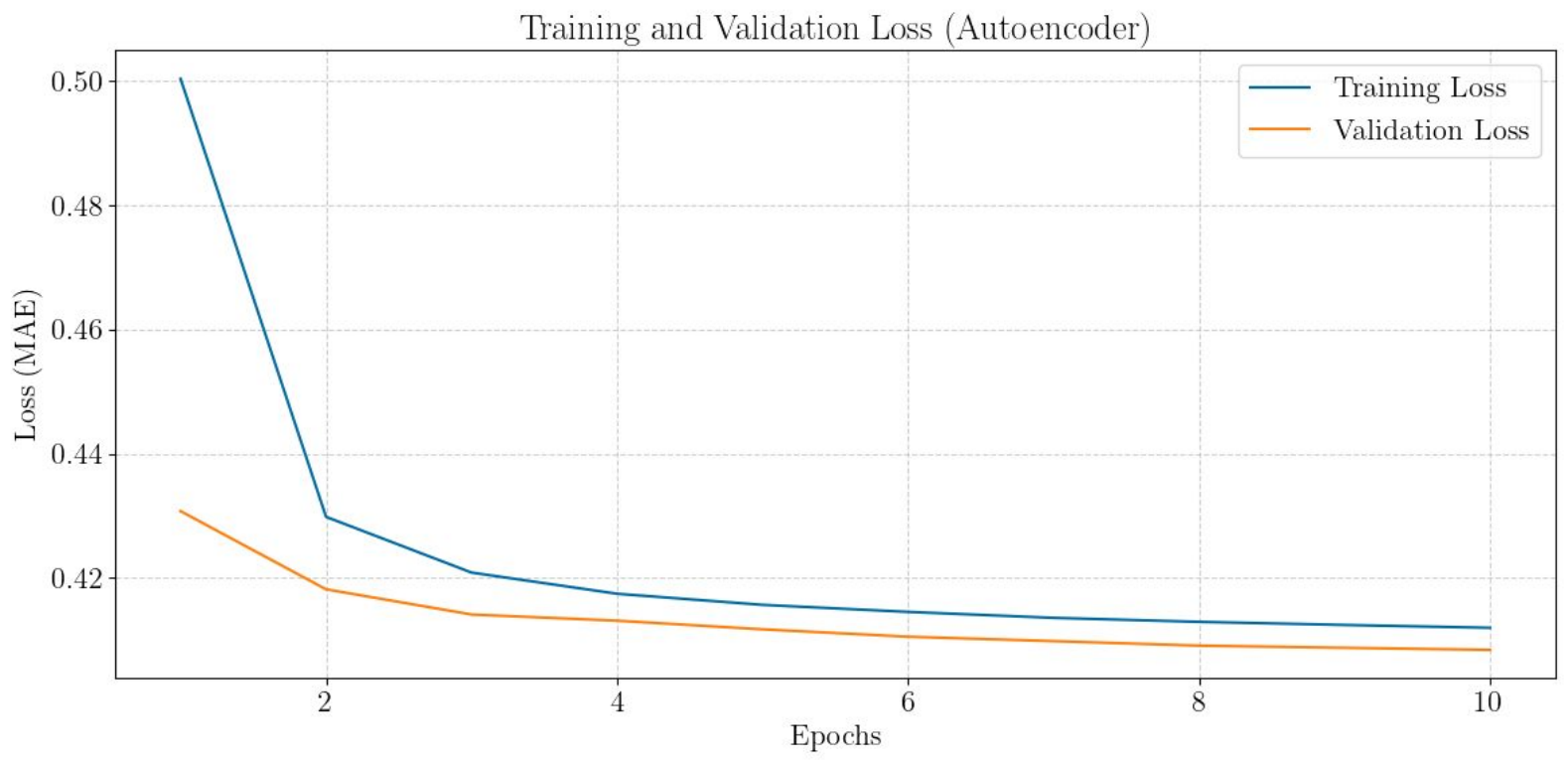
Attention

For Sensor Dropout

Baseline Loss - FD001



K-Cluster Loss - FD004



The Models - Performance

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To rank model performance we use the Spearman Coefficient, which is a measure of how realistic our fault detection is. A spearman coefficient of -1.0 indicates a perfect performance, while +1.0 indicates total failure. When it comes to idealistic test conditions (FD001), the Baseline model performs the best. For realistic test conditions (FD004), however, K-Cluster outperforms the rest. The attention model performs in the middle for all testing conditions.

Baseline

001 SP Coeff: -0.8

004 SP Coeff: -0.6

K-Cluster

001 SP Coeff: -0.3

004 SP Coeff: -0.7

Attention

001 SP Coeff: -0.7

004 SP Coeff: -0.6

Predicting Faults

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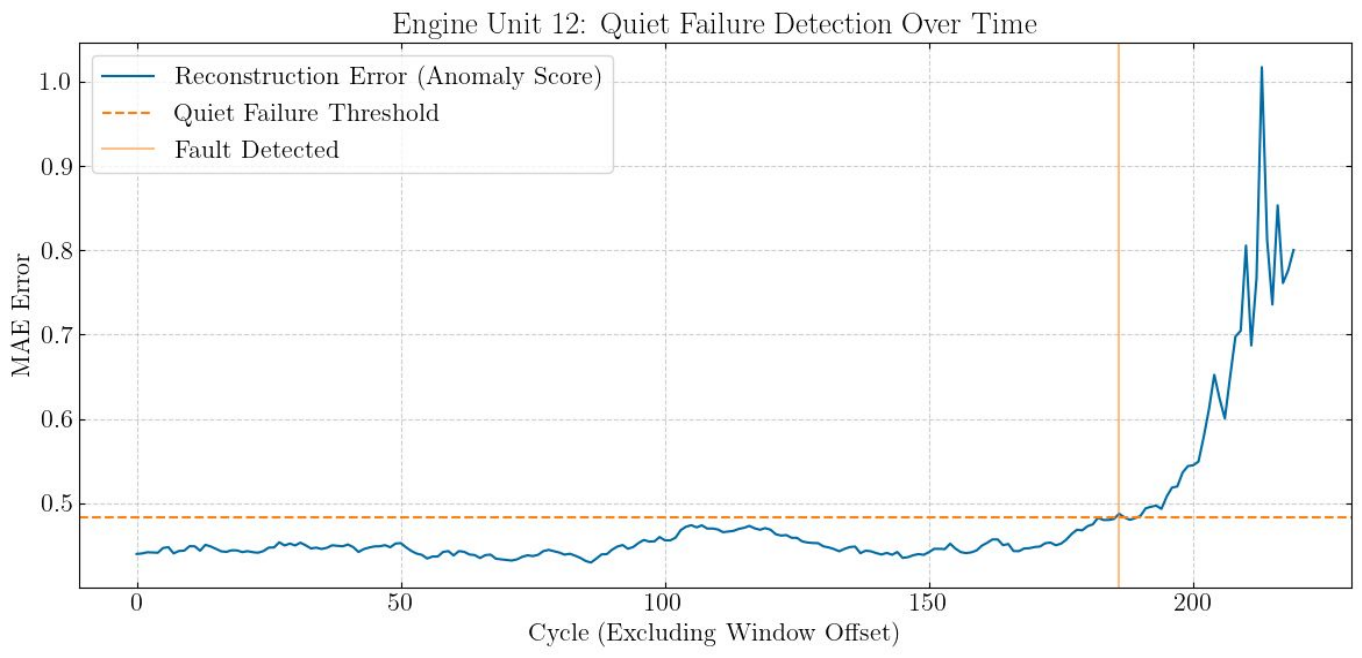

- Threshold is deemed as the 97.5th percentile of the average reconstruction loss on healthy data.
- Fault detected by autoencoder when exceeding threshold, if the fault exists.

Further Work

Informed Sensor Choice

Use the attention model to inform which sensors are more impactful and how to bias our models to those sensors. This could also help inform sensor placement for future engine design!

Statistical Thresholding

Select a more accurate methodology for identifying the failure threshold when predicting faults with the autoencoder

Better Tuning

Apply more rigorous hyperparameter tuning methods to our models to ensure that we are getting the best possible performance out of any given model

Thank you

Feel free to check out our GitHub Repo for more Information!